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Multi-scale planning model for robust urban drought response

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E-mail: mzaniolo@stanford.edu**Keywords:** urban drought security, water reuse, water desalination, water infrastructure adaptation, reinforcement learningSupplementary material for this article is available [online](#)**Abstract**

Increasingly severe droughts are straining municipal water resources and jeopardizing urban water security, but uncertainty in their duration, frequency, and intensity challenges drought planning and response. We develop the Drought Resilient Interscale Portfolio Planning model (DRIPP) to generate optimal planning responses to urban drought. DRIPP is a generalizable multi-scale framework for optimizing dynamic planning strategies of long-term infrastructure deployment and short-term drought response. It integrates climate and hydrological variability with high-fidelity representations of urban water distribution, available technology options, and demand reduction measures to yield robust and cost-effective water supply portfolios that are location-specific. We apply DRIPP in Santa Barbara, California to assess how least cost water supply portfolios vary under different drought scenarios and identify portfolios that are robust across drought scenarios. In Santa Barbara, we find that drought intensity, not duration or frequency, drives cost increases, reliability risk, and regret of overbuilding infrastructure. Under uncertain drought conditions, a diversified technology portfolio that includes both rapidly deployable, decentralized technologies alongside larger centralized technologies minimizes water supply cost while maintaining high robustness to climate uncertainty.

1. Introduction

The city of Santa Barbara built the Charles E. Mayer ocean desalination plant to address the water supply crisis caused by the late 1980s drought. After construction and commissioning, the plant was operated for four months (March–June, 1992), at which point abundant rainfall ended the drought and made desalinated water economically undesirable. The plant was modernized and reactivated in May, 2017 at an additional capital investment of \$72 million, nearly double the initial construction investment [1]. While desalinated water now provides rainfall-independent supply comprising 30% of Santa Barbara's water demand, the cost of producing and delivering this water is five-fold that of surface water [2]. These increased costs are reflected in household water bills and municipal energy demand.

Santa Barbara's history is not unique. In regions where water availability is highly variable and affected by climate change, water planners must make difficult infrastructure investment decisions that will affect water security, affordability, and equitable access decades into the future [3, 4]. Drought intensity, duration, and frequency, are expected to increase in many regions of the world [5], but how and how much is difficult to assess, and harder to predict then, for instance, average precipitation trends [6, 7]. When droughts abate, large expensive technologies appear to have been a bad investment. Some emerging technologies, like rapidly deployable small scale modular systems provide a more flexible alternative to centralized system deployment, but their per-unit cost of water treated is often higher [8]. Low-cost water supply portfolios that are robust in the face of changing climates require decision support tools that

explicitly assess the interactions between future climate uncertainty, hydrological variability, and technology attributes.

Choosing optimal sources, timing their implementation, and managing the portfolio of existing resources under uncertainty is too complex a problem for traditional decision-making methods like life-cycle cost-benefit analysis [9–11]. Instead, most academic water resources planning models incorporate decision-making under deep uncertainty and focus on developing strategies that are robust and adaptive to changing conditions. These approaches identify dynamic planning policies that respond to hydrological conditions as they unfold over time, with the aim of preventing over- and under-building [12–14]. For instance, these models use optimization to find low-cost robust water planning strategies in light of regional coordination opportunities [15, 16], implementation uncertainty [17], timing of infrastructural development [18, 19], drought characteristics [20], and preserving optionality for future infrastructure expansion [21, 22]. However, the water supply planning literature typically considers a limited number of hedging strategies [23] and infrastructure options [24] for water storage and diversion at the basin scale. This coarse spatial scale adequately represents climate uncertainty, but poorly represents water augmentation options at multiple scales, and typically aggregates cities into a single demand point.

Urban water literature, in contrast, has demonstrated that high-resolution, household-level modeling that explicitly considers distribution of water to end-users is necessary to accurately identify opportunities to integrate centralized and decentralized technologies [25–28], target demand-side management [29–31], and estimate the operational costs of urban or regional water supply networks [25, 32, 33]. Yet, these urban-level studies typically neglect climate uncertainty and hydrological variability, leading to static infrastructure portfolios that are costly and ineffective if water availability is highly variable.

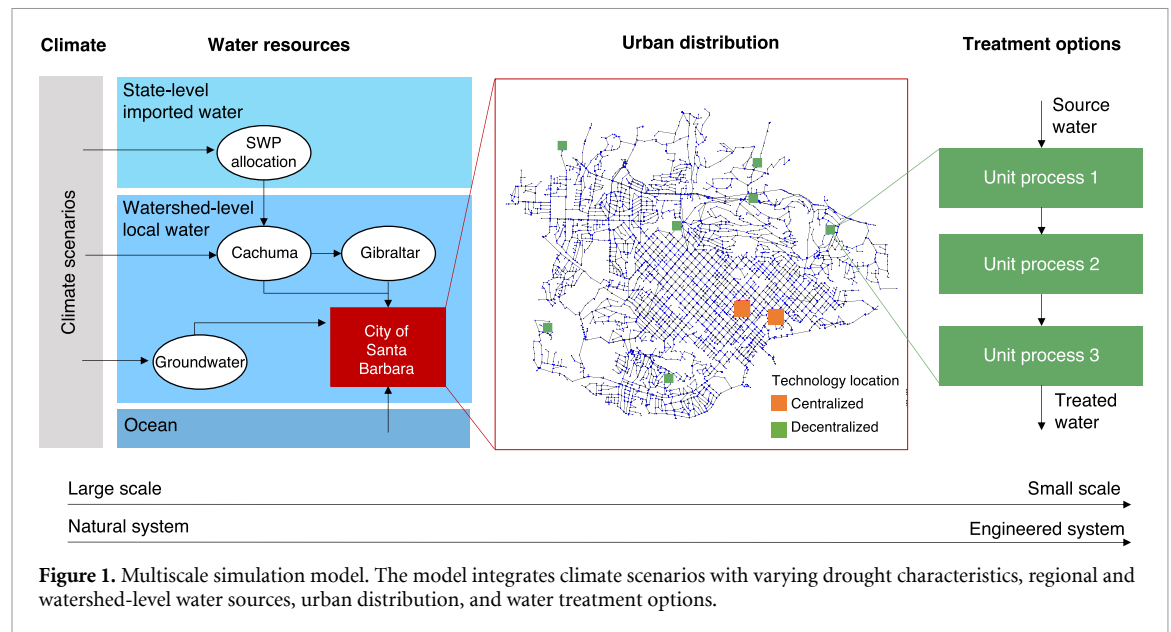
We need urban water planning frameworks that both respond to climate uncertainty at large scales, and represent urban demand and technologies at a high resolution. By integrating the state of the art in water resources planning and urban water literature, we can leverage the benefits of both scales, yielding to multi-scale, climate-robust solutions that include large to small-scale planning options and demand-side efforts. We introduce the Drought Resilient Inter-scale Portfolio Planning (DRIPP), a modeling framework that integrates regional and urban-scale water models to identify vulnerabilities to plausible future droughts and design robust solutions. State-of-the-art tools in these fields have never been brought together, in large part due to computational limitations. We overcome this challenge by drawing from recent advances in reinforcement learning methods

that couple universal approximators (e.g. neural networks, decision trees) with multi-objective evolutionary global parameter optimizers [34, 35]. We employ a decision tree-based method [36, 37] that allows us to optimally design water augmentation measures in combination with the hydrological and technological signposts that dynamically trigger their implementation across stochastic drought realizations.

We consider several non-traditional water supply sources and treatment options, and characterize them across multiple attributes including their levelized cost, time to deployment, and scalability. We create climate scenarios which reflect plausible changes in drought characteristics of frequency, duration, and intensity via synthetic streamflow generation [38]. We solve this optimization problem to find lowest-cost, robust planning strategies under these drought scenarios with bottom-up vulnerability analysis [12], in order to assess the relative influence of each drought attribute on planning strategies, water system costs, and robustness.

With this work, we expect to make two contributions. First, we create a framework that yields cost-effective and drought-robust water supply portfolios that respond dynamically to climate uncertainty and hydrological variability, and employ drought mitigation options at different scales and resolution. Second, we demonstrate that different types of drought require different planning strategies, and we analyse why certain portfolios are robust and low-cost in certain climates, but not others.

We apply DRIPP to the drought-stricken municipality of Santa Barbara, California. The state of California requires water utilities to demonstrate preparedness for a design drought with a duration of 5 years and an unprescribed intensity (i.e. magnitude of shortage) and frequency [39]. However, climate change projections show that frequency, duration, and intensity of droughts in California are likely to increase in magnitude and variability [6, 7], beyond the specified design drought characteristics. Our results show that the drought characteristics specified in climate scenarios directly influence urban water planning decisions and outcomes including cost, installed capacity, technology type, and system variables used to dynamically trigger decisions. In Santa Barbara—a system characterized by abundant surface storage with interannual carryover, and minimal groundwater—drought intensity, not duration or frequency, is the most influential attribute for determining the risk of supply deficit and the cost associated to the water portfolio. Additionally, rapidly deployed and easily scalable solutions like decentralized technology and curtailment measures are associated with higher annual cost in Santa Barbara, but are more robust across a wide variety of drought conditions, including out-of-sample climate scenarios that are not used in optimization.



2. Methods

In this work we apply DRIPP to test Santa Barbara's water system under climate scenarios displaying varying characteristics of drought frequency, duration, and intensity, and assess the relative influence of each drought attribute on optimal water system costs and planning strategies.

This section details and motivates the methods used in this work, namely the components of the multiscale simulation model, the optimization method, and the robustness analysis.

2.1. Multiscale simulation model

We represent Santa Barbara's water system with the multiscale simulation model in figure 1. This model integrates climate scenarios with varying drought characteristics, a conceptual water resources system model, an urban distribution model, and treatment train models for several treatment technologies. We present these elements below.

2.1.1. Identify drought characteristics that lead to system failure

Bottom-up vulnerability analysis employs a large sampling of plausible states of the world [40] to identify relevant changes in socio-hydrological conditions that produce consequential effects on the system [14, 41–44]. Identifying the conditions that lead to system vulnerability can direct planners to develop anticipatory solutions to prevent those outcomes.

Note that most bottom-up, vulnerability studies focus on changes in long-term average climate states, e.g. mean values of precipitation, temperature, streamflow [20]. However, short-term hydrological variability governed by changes in drought characteristics is also very influential for planning

and currently overlooked in bottom-up vulnerability literature. This study links long-term average climate change states to statistical patterns of short-term hydrological variability through uncertain drought characteristics with the goal of accurately capturing both long- and short-term vulnerability drivers.

In this work, a climate scenario comprises an ensemble of inflow time series to Santa Barbara's reservoirs with a specified average drought persistence P , intensity I , and frequency F . We design four climate scenarios: (1) Historical droughts: P , F , and I are comparable to recent droughts in Santa Barbara; (2) long mild droughts: double P , half I , and same F as historical droughts; (3) Short intense droughts: half P , double I and same F ; (4) frequent droughts: unaltered P and I , double F . P , I , and F are quantified using the standardized runoff index [45] (SRI) computed on streamflow time series (details in SI section 1). For each climate scenario, we generate an ensemble of 50 100 year streamflow time series using the synthetic drought generator presented in methods section 'Drought Generator' and illustrated in SI section 1.

Modeling different types of droughts allows us to systematically explore the space of plausible future vulnerability to droughts and design robust solutions revealing insights on their relative importance, and how they each influence planning strategies and costs. Lastly, we conduct an All droughts experiment that designs optimal planning policies on the ensemble of the four climate scenarios. By including these in the synthetic time series that the all drought policy is trained on, we ensure that the all drought policy is truly robust to all plausible future droughts—an outcome that a traditional synthetic time series approach using historical data or climate model projections would be unlikely to achieve.

As is common in bottom-up vulnerability literature, we generate climate scenarios using synthetic streamflow generation [40]. Most synthetic streamflow generation tools are designed for a specific application and allow the user to control a prespecified set of case-relevant streamflow properties, for instance AR parameters as informed by changes in precipitation and temperature [46], annual peak flow magnitude and total flow volume [47], and drought frequency and severity [20]. Borgomeo *et al* [38] proposed a streamflow generation method formulated as a multiobjective optimization problem where the objective functions represent the streamflow characteristics to be controlled. In absence of an existing synthetic generator that allows to control the drought attributes of interest for this study intensity, we employ Borgomeo's method and modify it to control the intensity, duration, and frequency of droughts in 100 years streamflow scenarios. The SI section 1 contains details and experimental design for the climate scenario generation, as well as selected plots of climate scenarios.

2.1.2. Water resources model

The water resources model captures the routing of water to the City of Santa Barbara from the surrounding sources, including inter-basin transfers from the California State Water Project (SWP), local water diverted from nearby reservoirs within the watershed, and desalinated ocean water. The monthly reservoir dynamics are described by the mass balance of their water volumes. According to the monthly time-step adopted in the model, the river reaches are modeled as plug-flow canals in which the velocity and direction of flow are constant. Groundwater abstraction is modeled as a constant 80 AF/m corresponding to historical sustainable average abstraction [2] due to data limitations and its small contribution to water supply (around 5%–7%). More information on the water resources model can be found in section 2 of the SI.

2.1.3. Urban water distribution model

The urban water distribution model simulates the distribution of water from surface water sources and water production technologies to individual nodes within the city through the water distribution network. The building blocks of this models are injection points connected to extra-urban water source, tanks, consumers, non-consumer junctions, pipes, valves and pumps. Simulating this model for a given water supply portfolio allows to estimate the energy intensity and cost of delivering water to end users for different centralized and decentralized technology configurations. More details on this model are presented in section 2 of the SI.

2.1.4. Water treatment models and water augmentation options

The portfolio of water augmentation options considered in this work includes a number of alternative water treatment technologies (water reuse, sea water desalination), modes of deployment (centralized and decentralized), and end-uses (potable and nonpotable). Additionally, we consider a demand reduction measure, curtailment, in which the city imposes mandatory water restrictions as a percentage of total demand [31, 48].

The treatment train models employed in this work estimate the capital and operational expenses of the considered treatment trains for a range of plant capacities. We estimate the costs of the considered technology options using the Water Technoeconomic Assessment Pipe-Parity Platform (WaterTAP3 [49]). WaterTAP3 is an open source platform that supports the simulation of a steady-state water treatment train and details its performance and costs and includes a variety of treatment unit processes configurations of treatment technologies, and system-level techno-economic assumptions. Section 3 of the SI reports details assumptions on additional technology attributes such as location, capacity, and time to deployment.

Table 1 summarizes the multi-attribute characterization of alternative water augmentation options and highlights their tradeoffs. In particular, decentralized deployment is typically associated with a higher unit cost (i.e. LCOW) than that of centralized technology, but it is also associated with faster and more scalable deployment. Non-potable reuse is less costly than potable reuse, but its end-use is limited to non-potable applications comprising only a fraction of the city's water demand. Curtailment is modeled as a water augmentation option whose capacity is equivalent to the curtailed demand with unit cost equivalent to the lost revenue incurred by the utility from reduced water sales. Water bills are high in a water-scarce municipality like Santa Barbara, which translates into high cost of curtailment measures, but their deployment is immediate. In addition, curtailment measures have been observed to induce a behavioral change for which demand does not rebound immediately to pre-drought levels after the mandate is lifted. The demand rebound effect has been empirically modeled as a survival process with a nearly 8 year half-life [50]. See the SI section 3 for more details on these assumptions.

2.1.5. Modeling choices and generalizability of the method

Note that we model a modified version of the current Santa Barbara water system. We consider a baseline water demand corresponding the city water use in 2013, preceding voluntary and mandated curtailment

Table 1. Multi-attribute characterization of the considered water augmentation options. The augmentation options (rows) are curtailment, centralized sea water desalination (C-SW), and centralized (C) and decentralized (D) potable reuse (PR) and non potable reuse (NPR). The attributes are the levelized cost of water (LCOW) calculated aggregating Capital and Operational expenses assuming a deployment of 200 acre feet/month (AF/m) for 40 years, the time to deployment (TTD), minimum and maximum capacity, and scalability. The data and assumptions used to populate this table are presented in section 3 of the SI.

	LCOW (\$/AF)	TTD (years)	Min capacity (AF/m)	Max capacity (AF/m)	Scalability
Curtailment	5998	0	75	375	5% through 25% of demand
C-SW	870	10	200	500	One plant with different capacity options
C-PR	804	5	100	500	One plant with different capacity options
C-NPR	733	4	100	200	One plant with different capacity options
D-PR	965	3	50	350	Up to 7 plants of 50 AF/m each
D-NPR	857	2	20	140	Up to 7 plants of 20 AF/m each

observed in the city during the drought of 2015–2016. We also consider a baseline water portfolio including only natural surface and groundwater sources, and excluding water technologies currently operational in the city, namely the Charles E. Meyer sea-water desalination plant. We made this choice with the aim to identify portfolio planning policies and analyze drought impacts independently from previous, potentially suboptimal, planning and curtailment measures taken by the city.

However, the DRIPP framework described in the next section can support analysis far beyond what is demonstrated by this case study. The user can swap the proposed water resources model for a different water system. Climate scenarios can be replaced with a different ensemble of synthetic scenarios probing vulnerability to other climate conditions. Water demand assumptions can be revisited to reflect changing demand and population growth. Costs, deployment, and capacity assumptions for the water augmentation options can be updated to capture alternative technological realities in different world regions, as well as to reflect technology innovation over time. Lastly, additional water augmentation measures can be considered, including increased water transfers, storage, enhanced regional coordination, and demand management options.

This multiscale simulation model is integrated in a simulation-optimization framework described in the next section.

2.2. The DRIPP framework

The DRIPP framework is a simulation-optimization approach that generates cost-optimal dynamic planning policies that incur no deficit on the range of climate conditions they are trained upon. In this work,

we design planning policies for each set of synthetic climate scenarios separately, generating, and then comparing, low-cost policies for historical, long and mild, intense, and frequent droughts. Lastly, we train a set of policies on all drought types conjunctively.

We formulate this cost-optimal urban water portfolio planning as a control problem [12] with flexible policy structure [37, 51, 52]. Figure 2(a) illustrates the method used to generate optimal planning policies π^* comprising the water augmentation decisions that minimize costs while achieving no unmet demand across the streamflow ensemble in each climate scenario. Decision policies are parameterized as an ensemble of decision trees that implement centralized, decentralized, and curtailment measures respectively in response to the state of water resources in the system. This state comprises several indicators that capture the hydrology, water storage, and technology over time. Table S2 and S3 in the SI contains a list of all indicators and actions considered in this work. Policies are optimized via an evolutionary simulation-optimization approach that progressively improves performance by iteratively generating and testing new policies, resulting in the optimal set of indicators, thresholds, actions, and functional structure that minimizes costs across the streamflow ensemble characterising a climate scenario. Studies typically fix one or more of these components *a priori* while optimizing the others, resulting in suboptimal empirical approximations. Here we leverage a recent method that allows us to optimize all these elements simultaneously [12], and we expand it to handle an ensemble of decision trees instead of a single tree. Our optimal decision policies therefore explicitly identify what information is used to trigger different augmentation options, improving the

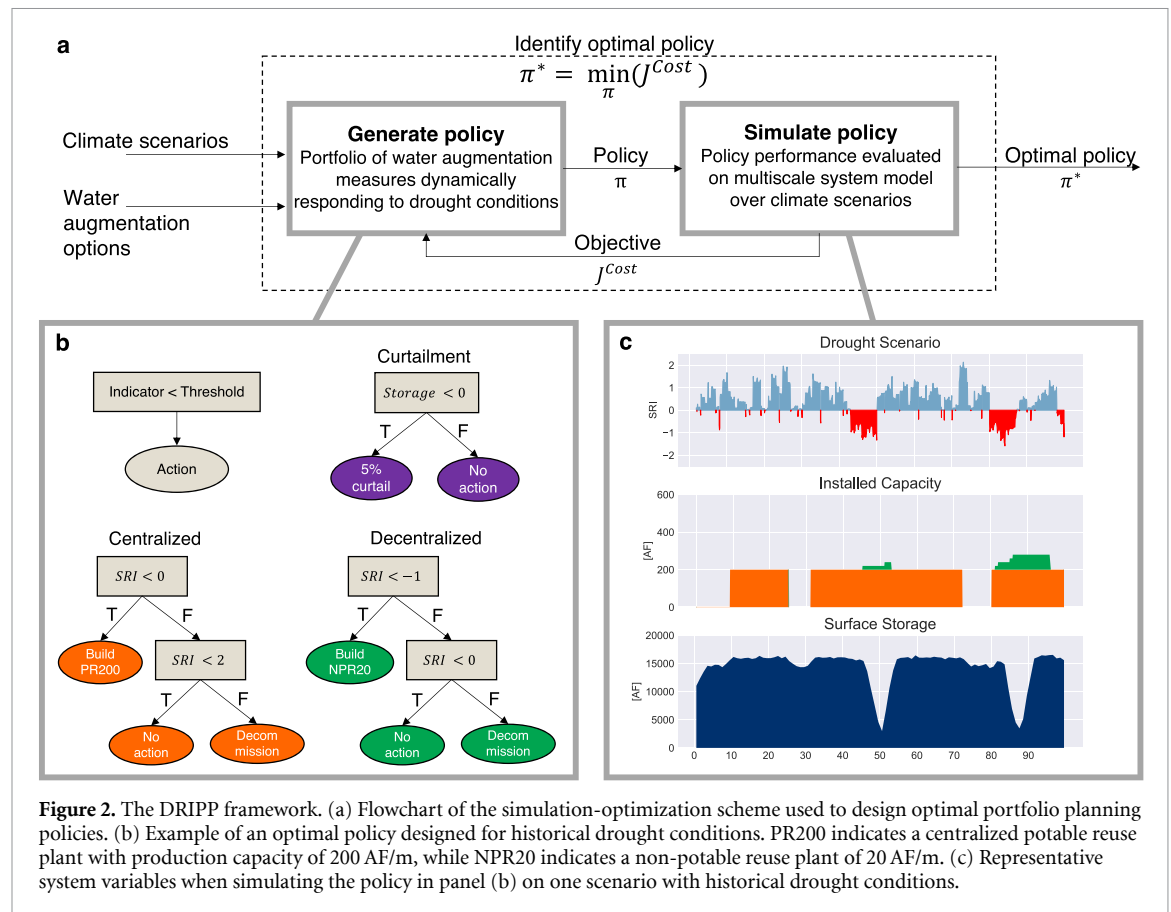


Figure 2. The DRIPP framework. (a) Flowchart of the simulation-optimization scheme used to design optimal portfolio planning policies. (b) Example of an optimal policy designed for historical drought conditions. PR200 indicates a centralized potable reuse plant with production capacity of 200 AF/m, while NPR20 indicates a non-potable reuse plant of 20 AF/m. (c) Representative system variables when simulating the policy in panel (b) on one scenario with historical drought conditions.

explainability of our results. See section 4 of the SI for the full problem formulation as well as the adopted experimental design.

Figure 2(b) shows an example of an optimal policy designed on 20 climate scenarios of 100 years with historical drought characteristics, and panel (c) displays representative system trajectories for the policy in panel (b) simulated on a 100 year historical drought scenario. Policy timeseries for other drought types and their analysis are presented in figure S6 of the SI.

2.2.1. Robustness analysis

Robustness is an important feature of planning policies in addressing climate uncertainty. A robust policy has low performance variability across climate scenarios, including out-of-sample scenarios that are not used in policy optimization, and is more likely to perform consistently in unknown future conditions.

In this paper we conduct two robustness tests. In the first analysis, we test policies optimized on all droughts across 30 out-of-sample climate scenarios of all drought types, and select a least-cost policy and a least-deficit policy based on their out-of sample performance. The comparison of these two policies reveals what attributes of a planning strategy make it robust to the effects of climate uncertainty on planning costs vs. unmet demand.

In the second analysis, we select policies trained on a given drought type (e.g. historical droughts), and

test them on a different drought type (e.g. short and intense droughts), quantifying the risk and regret of planning for the wrong drought. Our analysis quantifies the risk of underpreparing for future drought conditions and incurring deficits, as well as the regret of overspending on drought measures that may not be necessary.

Understanding the policy attributes and the climate conditions that contribute to policy robustness, or policy vulnerability, are key results of the bottom-up vulnerability analysis proposed in this work.

2.3. DRIPP within the landscape of existing water decision support tools

Several water system decision support models have been proposed over the years with advanced user interface, simulation capabilities, and representation of alternative stakeholders and sources of uncertainty (for a review, see [53]). DRIPP addresses four outstanding challenges that limit the utility of existing water system decision support tools. First, past approaches have relied exclusively on user's input data for technology options and cost parameters, which have typically been provided as point estimates rather than continuous functions. DRIPP integrates technology models that estimate the capital and operational costs of an array of traditional and non-traditional treatment trains as a function of design flow. Second, past models largely represent cities as single demand points, rather than a distributed network of water

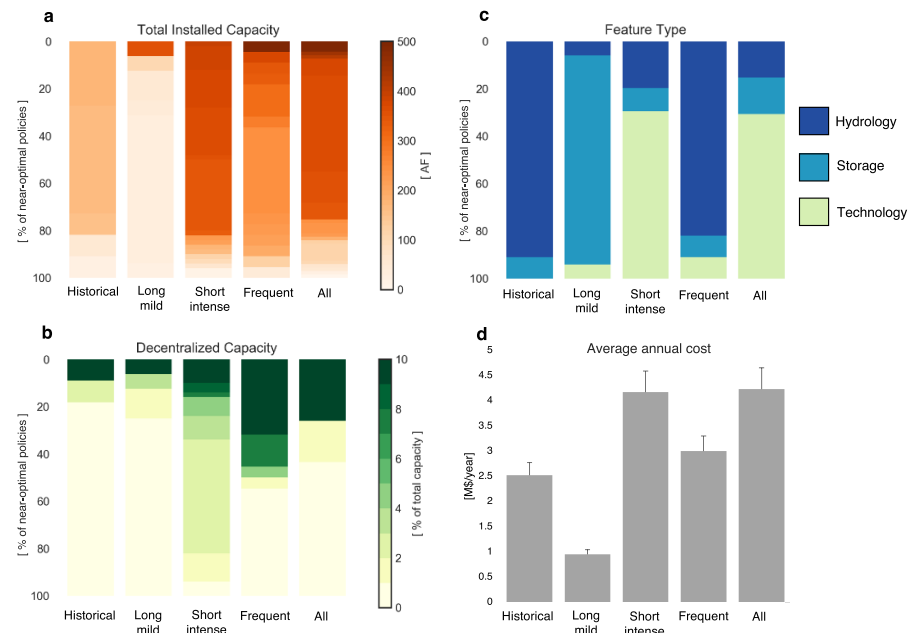


Figure 3. Cost-optimal planning policies for different drought types. (a)–(c) Characteristics of (near-)cost-optimal policies optimized for different drought types and averaged across 2000 simulation years. (a) Average installed capacity (b) Average fraction of decentralized capacity in the portfolio. (c) Category of indicator used to trigger capacity expansion. (d) Average annual cost of the cost-optimal drought planning policy designed for different drought types. The black bar indicates an additional 10% cost that constitutes the performance range achieved by near-optimal policies.

sources and users, thereby overlooking urban distribution costs, and missing opportunities to reduce distribution distance, and costs, with decentralized technology. DRIPP integrates water distribution network models to estimate the cost of delivering water to end users within a city, which often represents a significant fraction of a utility's expenses and may vary with the portfolio of technology installed [25]. Third, in existing methods, action deployment is triggered exclusively by looking at storage-based indicators [53], while DRIPP automatically selects a set of relevant indicators from a multidimensional dataset of indicators representing the state of the climate, hydrology, and technology at various time aggregation. Fourth, vulnerability studies have largely focused on changes in long-term average climate states, (e.g. mean values of precipitation, temperature, stream-flow). DRIPP more accurately captures vulnerability to extreme events by considering droughts with varying characteristics of duration, frequency, and intensity. This addition enables important new insights on the interactions between drought characteristics and technology attributes that are not possible to identify using existing modeling approaches. Specifically, it allows us to identify that drought intensity, rather than frequency or duration, drives vulnerability in our case study, which has important policy implications given that that the design drought in California is currently based on drought duration. Additionally, it allows us to identify the conditions under which small-scale decentralized technologies are useful. Our results highlight that policies that are robust to all

drought characteristics rely on decentralized technology: while it does not comprise a large fraction of installed capacity, it reduces system costs by reducing the need to overbuild large centralized capacity.

However, we believe there are several opportunities to pair DRIPP with existing water decision support tools with advanced user interfaces. Planning pathways designed with DRIPP's multi-scale optimization approach can be evaluated and contrasted via water system models with a user-friendly graphical interface [54, 55], integration with geographic information system (GIS) [56], and models that estimate user preferences beyond cost [53, 57].

3. Results

3.1. Optimal urban portfolio planning policies for different climate scenarios

Figure 3 shows the impact of climate scenarios with varying drought characteristics on optimal urban drought planning decisions in terms of deployed technology (panels (a) and (b)), indicators used to trigger deployment (panel (c)), and annual water supply costs (panel (d)). We design cost-optimal planning policies that yield no deficit for each drought type independently and for all droughts in order to capture how drought characteristics influence optimal planning decisions and outcomes.

Panel (a) shows the average capacity deployed by cost-optimal and near-optimal policies (i.e. within 10% of lowest supply costs) designed for different climate scenarios. The intensity of the color indicates

the magnitude of capacity deployed by different (near-)cost-optimal policies. For example, the first bar in figure 3(a) indicates that 80% of the near-optimal policies designed for Historical drought conditions deploy a total water supply capacity of 150 AF/month or more. The remaining 20% of cost-optimal policies deploy less capacity, but incur in similar costs. This indicates that while some planning strategies are lower cost than others, planners can choose from a range of alternative policies without increasing total costs more than 10%. In particular, the ensemble of low-cost policies contains both low and high deployment strategies. Low deployment strategies select a large fraction of decentralized technology (panel (b)) and curtailment measures that come online just-in-time in response to depleting water reserves, and scale production to water needs, with higher unit cost. Conversely, high deployment strategies select centralized technology with low unit cost and high inertia (i.e. poor scalability and high time to deployment) that remain online when not strictly necessary.

In contrast to the historical climate scenario, planning for a climate with long mild droughts requires a relatively low installed capacity, as surface water storage depletes slowly and allows time for technology to come online when needed. Conversely, short intense droughts develop abruptly, leaving water planners little time to respond to early signs of drought and requiring large redundancy in installed capacity to ensure enough water production. Low-cost planning policies for all droughts display a capacity distribution similar to short intense droughts.

Panel (b) shows the average fraction of decentralized technology deployed in the portfolio. Decentralized capacity is typically deployed dynamically at the onset of a drought, as surface water storage is depleting and the city needs quick-to-deploy temporary augmentation measures. As a result, while the fraction of decentralized capacity may appear low when averaged across the simulation, decentralized measures are essential to prevent shortages in critical conditions. This is why decentralized deployment is highest in the frequent drought climate scenarios.

The drought type also determines the type of indicators used to trigger capacity expansion (panel (c)). (Near-) optimal policies designed for long mild droughts deploy capacity in response to storage indicators, which capture the slow decline in water resources in lakes and reservoirs. Planning policies in historical and frequent drought climates mostly respond to hydrological indicators (e.g. drought index) that might reveal the onset of a drought before it has a measurable impact on storage levels. Short intense drought policies mainly deploy capacity in response to technology indicators, e.g. current installed capacity. The onset of these droughts is so abrupt that new technology cannot be deployed dynamically in response to unfolding

drought conditions. Conversely, it is deployed if the current installed capacity would not be adequate to buffer an incoming drought. Lastly, policies designed to be robust across all drought types present a similar indicators as Short intense droughts.

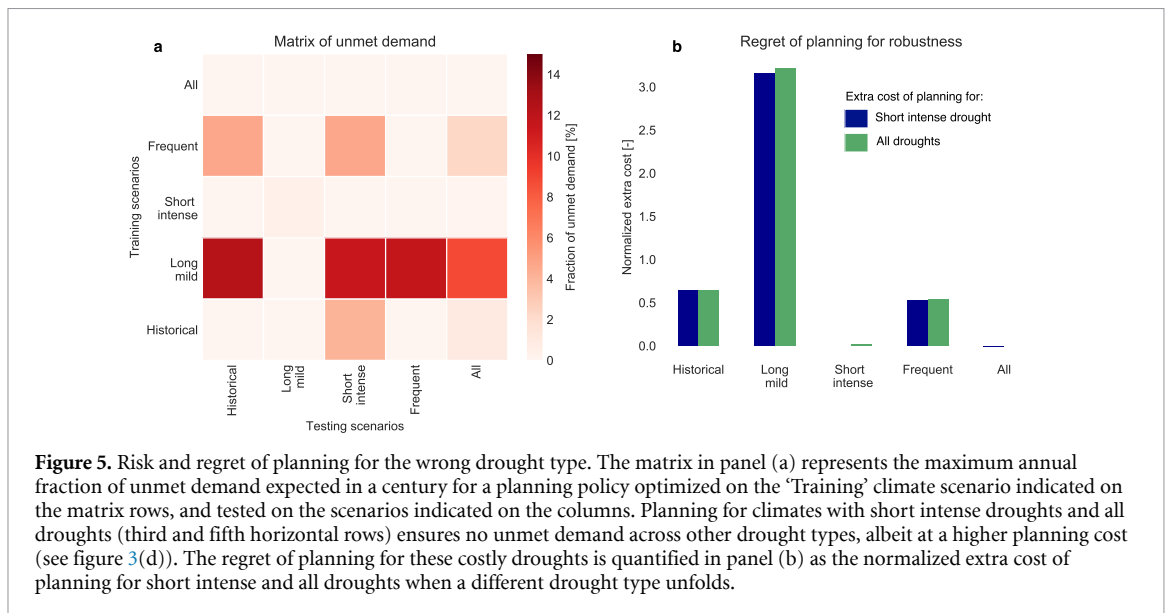
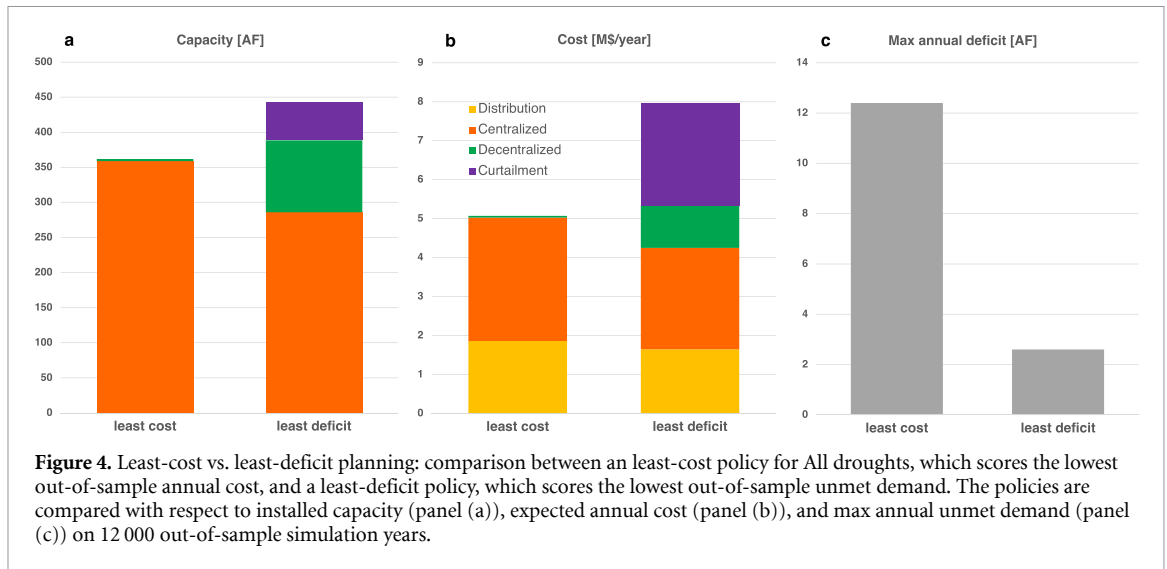
The difference in planning strategies leads to very different annual costs (panel (d)). Planning for short intense droughts and all droughts would cost the city of Santa Barbara over 4 M\$/year, about four times the cost of planning for long mild droughts. Planning for historical and frequent droughts results in about 2.5 M\$/year and 3 M\$/year respectively.

3.2. Ensuring robust planning across all droughts

This section presents the results of the first robustness analysis performed in this work, and compares the attributes of two policies trained on all droughts and achieving, respectively, lowest out-of-sample costs (i.e. least-cost policy) and lowest out-of-sample unmet demand (least-deficit policy). Capacity, cost, and maximum annual unmet demand of least-cost and least-deficit policies are shown in figure 4. The least-deficit policy installs a higher average capacity than the least-cost policy (panel (a)), indicating that including more redundancy in the system enables high robustness to deficit. Interestingly, however, the redundancy is not achieved by merely oversizing the least-cost technology portfolio, but by enhancing its flexibility and creating a more diversified portfolio of measures that includes lower centralized capacity and a significant portion of decentralized capacity and curtailment measures. Given the higher unit cost of these drought measures, the cost of the least-deficit portfolio is 60% higher than the least-cost, indicating that designing for robustness to water shortage comes at a significant cost for the system (panel (b)). The water distribution network model estimates the cost of delivering water to users. Deploying decentralized technology is shown to reduce the distance between the water source and the water users, thereby reducing distribution costs. However, this difference is small in Santa Barbara's mostly flat distribution network, and certainly not enough to offset the higher unit cost of decentralized measures. On the other hand, robust least-cost planning exposes the system to shortage risks, in this analysis quantified as six times larger for the least-cost policy than for the least-deficit policy (panel (c)).

3.3. Risk and regret of planning for the wrong drought type

This section presents the results of the second robustness analysis investigating the effects of planning for the wrong drought. Drought planning in an uncertain climate presents both the risk of incurring water deficit through under-preparation for a severe drought and the regret of overspending on unnecessary drought measures. Figure 5 quantifies the risk and regret of planning for the wrong drought type



in the city of Santa Barbara. The matrix in panel (a) shows the maximum annual unmet demand expected in a 100 year time period for a policy optimized on drought conditions in one climate scenario (row) and tested across other climate scenarios (columns). Planning for a climate with long mild droughts (second row) produces large unmet demand across all other climates. Short and intense droughts are the most challenging (middle column), producing unmet demand on policies optimized for all other climate scenarios. Planning for short intense droughts minimizes the risk of deficit if a different drought occurs (third row), equivalent to planning for all drought types (top row). However, planning for short intense and all droughts is expensive (figure 3(d)). Figure 5(b) shows the extra costs, or regret, of planning for Short intense and all droughts, in climate scenarios without intense droughts. For example, the regret R of planning for an expensive Short intense drought when a long mild drought

unfolds is here computed as $R = (C^{SI} - C^{LM})/C^{LM}$ where C^{SI} is the cost of planning for a short intense drought, and C^{LM} is the cost of planning for a long mild drought. In this case the regret is significant, leading to up to a three-fold increase in cost.

4. Discussion

This work presents DRIPP, a framework for dynamic and drought-responsive urban water portfolio planning demonstrated for the city of Santa Barbara, California. Our results underscore that high-resolution representation of technology, climate, and system modeling are fundamental building blocks of effective urban drought planning under a changing climate. According to the middle-range theory in sustainability [58], this case study bears relevance for many drought-prone municipalities characterized by low annual precipitation and large dependence on surface storage. Additionally, DRIPP is easily

transferable to different case studies upon customizing the hydrological and urban distribution models. Below, we detail their importance and implications for planning, and we highlight areas where further research is needed.

4.1. Technology matters

The 20th century approach to ensuring robustness of a water system is through redundancy: oversizing traditional centralized infrastructure by a safety factor [8, 59]. Our results for Santa Barbara highlight suboptimal outcomes of redundancy planning under growing climate and demand uncertainty. Instead, high robustness to water shortages in this case study is achieved through a diversified portfolio of water augmentation measures that relies on cost-effective centralized technology for baseload and on rapidly deployed, modular technologies that dynamically respond to droughts. However, flexibility and robustness increase costs by 60%. Further research to reduce costs of small scale, decentralized technology is critical to lower cost of portfolio diversification in a changing climate. Additionally, research in small-scale water treatment technology could critically enhance urban drought resilience while limiting overbuilding.

4.2. Climate matters

The drought characteristics of the future climate we prepare for significantly impact local planning decisions. In Santa Barbara, the optimal planning strategy to address long and mild droughts is radically different than the optimal strategy to address short intense droughts. Our results show that drought intensity, not duration or frequency, drives both risk of deficit during a drought we are not prepared for, and the regret of overspending on unnecessary drought measures. These findings underscore the value in advancing our ability to predict the characteristics of upcoming droughts. While previous work has explored how to predict the onset of an upcoming drought [60, 61], further research is needed to advance our ability to predict the persistence, frequency, and especially intensity of future droughts.

4.3. System representation matters

The problem of urban water security encompasses multiple scales from climate, watershed, city, and ultimately end users. Maintaining high spatial, temporal, technology, and climate fidelity across scales yields more realistic and implementable water supply portfolios. Additionally, high-fidelity, multi-scale and multi-attribute models are more likely to be trusted and adopted by diverse stakeholders because the model fidelity matches or exceeds that used to simulate their sub-systems. While accurate system representation is important, the strictly technical DRIPP model does not capture elements of

social preferences and concerns for alternative water treatment technologies that are rarely captured in computational water system models. For instance, recent social science literature has shown that sea-water desalination plants are often perceived by the public as very energy intensive and raise concerns of excessive greenhouse gas emissions [2, 62], and decentralized wastewater treatment systems might be unpopular due to water quality concerns [63]. A planning tool like DRIPP should be accompanied by holistic planning processes that better capture social and political preferences that inform influence a water planner's decision.

4.4. Implications for drought planning in California

Our findings have implications for policymakers, municipal water planners, and technology providers in the state of California and beyond. Currently, the state of California requires water utilities to ensure preparedness for a design drought of 5 years and an unprescribed intensity and frequency. Our findings show that intensity is the primary drought attribute that influences planning decisions, and determines the tradeoff between risk and regret of planning for the wrong drought. While this study focuses on the city of Santa Barbara, our findings and recommendations might be generalized to similar drought prone coastal cities in California. We recommend policy makers in the state of California include drought intensity in the design drought. We advise municipal water planners in California to use DRIPP to assess the value of a diversified water portfolio that includes centralized technology for a cost-effective baseload and rapidly deployable, distributed technologies as a tool for augmenting robustness under drought uncertainty. Lastly, we recommend that technology providers advance research in decentralized and modular technologies with the aim of enabling municipalities to diversify their water portfolio cost effectively.

Data availability statement

The hydrological data used in this study is publicly available on the USGS National Water Dashboard <https://dashboard.waterdata.usgs.gov/app/nwd/?region=lower48&aoi=default>. The record of SWP allocation is available at the SWP website <https://water.ca.gov/Programs/State-Water-Project>. Technological data and assumptions used in this project are the default assumptions built in the WaterTAP3 models. Water reservoirs geomorphological data was acquired from the US Bureau of Reclamation Projects and Facilities Database www.usbr.gov/projects/. Lastly, the City of Santa Barbara's water distribution network is considered sensitive information for the city, and was shared with the authors confidentially.

Code availability

The WaterTAP3 framework used to estimate energy requirements and costs of alternative treatment train technologies is available at <https://github.com/NREL/WaterTAP3>. The WNTR framework used to simulate the urban distribution network is available at <https://github.com/USEPA/WNTR>. The planning policy optimization is based on a wrapper of the open PTREEOPT code available at <https://github.com/jdherman/ptreeopt>. Case study-specific models and functions are available on the author's GitHub (<https://github.com/m-zaniolo/DRIPP>).

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